Problem A. Shortest Path 3

Time Limit 2000 ms

Problem Statement

You are given a simple connected undirected graph with N vertices and M edges. Each vertex i $(1 \leq i \leq N)$ has a weight A_i. Each edge j $(1 \leq j \leq M)$ connects vertices U_j and V_j bidirectionally and has a weight B_j.

The weight of a path in this graph is defined as the sum of the weights of the vertices and edges that appear on the path.

For each $i = 2, 3, \ldots, N$, solve the following problem:

- Find the minimum weight of a path from vertex 1 to vertex i.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $N - 1 \leq M \leq 2 \times 10^5$
- $1 \leq U_j < V_j \leq N$
- $(U_i, V_i) \neq (U_j, V_j)$ if $i \neq j$.
- The graph is connected.
- $0 \leq A_i \leq 10^9$
- $0 \leq B_j \leq 10^9$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N\ M \\
A_1\ A_2\ \ldots\ A_N \\
U_1\ V_1\ B_1 \\
U_2\ V_2\ B_2 \\
\vdots \\
U_M\ V_M\ B_M
\end{array}
$$

Output

Print the answers for $i = 2, 3, \ldots, N$ in a single line, separated by spaces.

Sample 1

Input	Output
3 3 1 2 3 1 2 1 1 3 6 2 3 2	4 9

Consider the paths from vertex 1 to vertex 2. The weight of the path $1 \to 2$ is $A_1 + B_1 + A_2 = 1 + 1 + 2 = 4$, and the weight of the path $1 \to 3 \to 2$ is $A_1 + B_2 + A_3 + B_3 + A_2 = 1 + 6 + 3 + 2 + 2 = 14$. The minimum weight is 4.

Consider the paths from vertex 1 to vertex 3. The weight of the path $1 \to 3$ is $A_1 + B_2 + A_3 = 1 + 6 + 3 = 10$, and the weight of the path $1 \to 2 \to 3$ is $A_1 + B_1 + A_2 + B_3 + A_3 = 1 + 1 + 2 + 2 + 3 = 9$. The minimum weight is 9.

Sample 2

Input	Output
2 1 0 1 1 2 3	4

Sample 3

Input	Output
5 8 928448202 994752369 906965437 942744902 907560126 2 5 975090662 1 2 908843627 1 5 969061140 3 4 964249326 2 3 957690728 2 4 942986477 4 5 948404113 1 3 988716403	2832044198 2824130042 4696218483 2805069468

Note that the answers may not fit in a 32-bit integer.

Problem B. Come Back Quickly

Time Limit 3000 ms

Problem Statement

In the Republic of AtCoder, there are N towns numbered 1 through N and M roads numbered 1 through M.
Road i is a one-way road from Town A_i to Town B_i, and it takes C_i minutes to go through.
It is possible that $A_i = B_i$, and there may be multiple roads connecting the same pair of towns.
Takahashi is thinking about taking a walk in the country. We will call a walk **valid** when it goes through one or more roads and returns to the town it starts at.
For each town, determine whether there is a valid walk that starts at that town.
Additionally, if the answer is yes, find the minimum time such a walk requires.

Constraints

- $1 \le N \le 2000$
- $1 \le M \le 2000$
- $1 \le A_i \le N$
- $1 \le B_i \le N$
- $1 \le C_i \le 10^5$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ M \\
A_1 \ B_1 \ C_1 \\
A_2 \ B_2 \ C_2 \\
A_3 \ B_3 \ C_3 \\
\quad \vdots \\
A_M \ B_M \ C_M
\end{array}
$$

Output

Print N lines. The i-th line $(1 \leq i \leq N)$ should contain the following:

- if there is a valid walk that starts at Town i, the minimum time required by such a walk;
- otherwise, `-1`.

Sample 1

Input	Output
4 4 1 2 5 2 3 10 3 1 15 4 3 20	30 30 30 -1

By Roads $1, 2, 3$, Towns $1, 2, 3$ forms a ring that takes 30 minutes to go around.
From Town 4, we can go to Towns $1, 2, 3$, but then we cannot return to Town 4.

Sample 2

Input	Output
4 6 1 2 5 1 3 10 2 4 5 3 4 10 4 1 10 1 1 10	10 20 30 20

There may be a road such that $A_i = B_i$.
Here, we can use just Road 6 to depart from Town 1 and return to that town.

Sample 3

Input	Output
4 7 1 2 10 2 3 30 1 4 15 3 4 25 3 4 20 4 3 20 4 3 30	-1 -1 40 40

Note that there may be multiple roads connecting the same pair of towns.

Problem C. Super Takahashi Bros.

Time Limit 2000 ms

Problem Statement

Takahashi is playing a game.

The game consists of N stages numbered $1, 2, \ldots, N$. Initially, only stage 1 can be played.

For each stage i ($1 \leq i \leq N-1$) that can be played, you can perform one of the following two actions at stage i:

- Spend A_i seconds to clear stage i. This allows you to play stage $i+1$.
- Spend B_i seconds to clear stage i. This allows you to play stage X_i.

Ignoring the times other than the time spent to clear the stages, how many seconds will it take at the minimum to be able to play stage N?

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq A_i, B_i \leq 10^9$
- $1 \leq X_i \leq N$
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$\begin{array}{l} N \\ A_1 \ B_1 \ X_1 \\ A_2 \ B_2 \ X_2 \\ \vdots \\ A_{N-1} \ B_{N-1} \ X_{N-1} \end{array}$$

Output

Print the answer.

Sample 1

Input	Output
5 100 200 3 50 10 1 100 200 5 150 1 2	350

By acting as follows, you will be allowed to play stage 5 in 350 seconds.

- Spend 100 seconds to clear stage 1, which allows you to play stage 2.
- Spend 50 seconds to clear stage 2, which allows you to play stage 3.
- Spend 200 seconds to clear stage 3, which allows you to play stage 5.

Sample 2

Input	Output
10 1000 10 9 1000 10 10 1000 10 2 1000 10 3 1000 10 4 1000 10 5 1000 10 6 1000 10 7 1000 10 8	90

Sample 3

Input	Output
6 1000000000 1000000000 1 1000000000 1000000000 1 1000000000 1000000000 1 1000000000 1000000000 1 1000000000 1000000000 1	5000000000

Problem D. At Most 3 (Judge ver.)

Time Limit 2000 ms

Problem Statement

There are N weights called Weight 1, Weight 2, $\ldots$, Weight N. Weight i has a mass of A_i.
Let us say a positive integer n is a **good integer** if the following condition is satisfied:

- We can choose **at most** 3 different weights so that they have a total mass of n.

How many positive integers less than or equal to W are good integers?

Constraints

- $1 \leq N \leq 300$
- $1 \leq W \leq 10^6$
- $1 \leq A_i \leq 10^6$
- All values in input are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \ W \\
A_1 \ A_2 \ \ldots \ A_N
\end{array}
$$

Output

Print the answer.

Sample 1

Input	Output
2 10 1 3	3

If we choose only Weight 1, it has a total mass of 1, so 1 is a good integer.
If we choose only Weight 2, it has a total mass of 3, so 3 is a good integer.
If we choose Weights 1 and 2, they have a total mass of 4, so 4 is a good integer.
No other integer is a good integer. Also, all of 1, 3, and 4 are integers less than or equal to W. Therefore, the answer is 3.

Sample 2

Input	Output
2 1 2 3	0

There are no good integers less than or equal to W.

Sample 3

Input	Output
4 12 3 3 3 3	3

There are 3 good integers: 3, 6, and 9.
For example, if we choose Weights 1, 2, and 3, they have a total mass of 9, so 9 is a good integer.
Note that 12 is **not** a good integer.

Sample 4

Input	Output
7 251 202 20 5 1 4 2 100	48

Problem E. Takahashi the Wall Breaker

Time Limit 2000 ms

Problem Statement

Takahashi is about to go buy eel at a fish shop.

The town where he lives is divided into a grid of H rows and W columns. Each cell is either a road or a wall.
Let us denote the cell at the i-th row from the top $(1 \leq i \leq H)$ and the j-th column from the left $(1 \leq j \leq W)$ as cell (i, j).
Information about each cell is given by H strings $S_1, S_2, \ldots, S_H$, each of length W. Specifically, if the j-th character of S_i $(1 \leq i \leq H, 1 \leq j \leq W)$ is `.`, cell (i, j) is a road; if it is `#`, cell (i, j) is a wall.

He can repeatedly perform the following two types of actions in any order:

- Move to an adjacent cell (up, down, left, or right) that is within the town and is a road.
- Choose one of the four directions (up, down, left, or right) and perform a **front kick** in that direction.
  When he performs a front kick, for each of the cells at most 2 steps away in that direction from the cell he is currently in, if that cell is a wall, it becomes a road.
  If some of the cells at most 2 steps away are outside the town, a front kick can still be performed, but anything outside the town does not change.

He starts in cell (A, B), and he wants to move to the fish shop in cell (C, D).
It is guaranteed that both the cell where he starts and the cell with the fish shop are roads.
Find the minimum **number of front kicks** he needs in order to reach the fish shop.

Constraints

- $1 \leq H \leq 1000$
- $1 \leq W \leq 1000$
- Each S_i is a string of length W consisting of `.` and `#`.
- $1 \leq A, C \leq H$
- $1 \leq B, D \leq W$
- $(A, B) \neq (C, D)$
- H, W, A, B, C, and D are integers.

- The cell where Takahashi starts and the cell with the fish shop are roads.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
H \ W \\
S_1 \\
S_2 \\
\vdots \\
S_H \\
A \ B \ C \ D
\end{array}
$$

Output

Print the minimum number of front kicks needed for Takahashi to reach the fish shop.

Sample 1

Input	Output

`10 10`
`..........`
`#########.`
`#.......#.`
`#..####.#.`
`##....#.#.`
`#####.#.#.`
`.##.#.#.#.`
`###.#.#.#.`
`###.#.#.#.`
`#.....#...`
`1 1 7 1`
	`1`

Takahashi starts in cell $(1, 1)$.
By repeatedly moving to adjacent road cells, he can reach cell $(7, 4)$.
If he performs a front kick to the left from cell $(7, 4)$, cells $(7, 3)$ and $(7, 2)$ turn from walls to roads.
Then, by continuing to move through road cells (including those that have become roads), he can reach the fish shop in cell $(7, 1)$.

In this case, the number of front kicks performed is 1, and it is impossible to reach the fish shop without performing any front kicks, so print 1.

Sample 2

Input	Output

`2 2`
`.#`
`#.`
`1 1 2 2`
	`1`

Takahashi starts in cell $(1, 1)$.
When he performs a front kick to the right, cell $(1, 2)$ turns from a wall to a road.
The cell two steps to the right of $(1, 1)$ is outside the town, so it does not change.
Then, he can move to cell $(1, 2)$ and then to the fish shop in cell $(2, 2)$.

In this case, the number of front kicks performed is 1, and it is impossible to reach the fish shop without performing any front kicks, so print 1.

Sample 3

Input	Output

`1 3`
`.#.`
`1 1 1 3`
	`1`

When performing a front kick, it is fine if the fish shop's cell is within the cells that could be turned into a road. Specifically, the fish shop's cell is a road from the beginning, so it remains unchanged; particularly, the shop is not destroyed by the front kick.

Sample 4

Input	Output
20 20 #################### ##...##....###...### #.....#.....#.....## #..#..#..#..#..#..## #..#..#....##..##### #.....#.....#..##### #.....#..#..#..#..## #..#..#.....#.....## #..#..#....###...### #################### #################### ##..#..##...###...## ##..#..#.....#.....# ##..#..#..#..#..#..# ##..#..#..#..#..#..# ##.....#..#..#..#..# ###....#..#..#..#..# #####..#.....#.....# #####..##...###...## #################### 3 3 18 18	3

Problem F. Puddles

Time Limit 2000 ms

Problem Statement

There is a grid with H rows and W columns.
The cell at the i-th row from the top and j-th column from the left is painted black if $S_{i,j}$ is `#`, and white if $S_{i,j}$ is `.`.

Among the four-directionally connected regions consisting of white cells, find the number of those that are surrounded by black cells.

Below is a more formal statement.

We denote the cell at the i-th row from the top and j-th column from the left as cell (i, j).
Two cells (i, j) and (i', j') are said to be adjacent if and only if $|i - i'| + |j - j'| = 1$.
A set C of white cells is said to be connected if and only if, for any two cells c and c' in C, it is possible to move from c to c' by repeatedly moving to an adjacent cell in C.
A non-empty connected set of white cells that is maximal is called a connected component of white cells.
Find the number of connected components of white cells that do not contain any cell on the outermost border of the grid (that is, in row 1, row H, column 1, or column W).

Constraints

- $3 \leq H, W \leq 10^3$
- H and W are integers.
- $S_{i,j}$ is `#` or `.`.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
H \ \ W \\
S_{1,1}S_{1,2}\ldots S_{1,W} \\
\vdots \\
S_{H,1}S_{H,2}\ldots S_{H,W}
\end{array}
$$

Output

Output the answer.

Sample 1

Input	Output
5 15 ##########..### #...#######.### ####....###..## ######.######## ########....###	2

There are two such regions: the region consisting of the three cells in row 2 from the top and columns 2, 3, 4 from the left, and the region consisting of a total of five cells: columns 5, 6, 7, 8 from the left in row 3 from the top and column 7 from the left in row 4 from the top.

Sample 2

Input	Output
10 22 ###################### ####.################# ###...################ ##.###.##.....######## ##.....##.####.####### .######.#......#.....# .######.#.####.#.##### #########.....##.##### ################.##### ################.....#	4

Problem G. Poem Online Judge

Time Limit 2000 ms

Problem Statement

Poem Online Judge (POJ) is an online judge that gives scores to submitted strings.
There were N submissions to POJ. In the i-th earliest submission, string S_i was submitted, and a score of T_i was given. (The same string may have been submitted multiple times.)
Note that POJ **may not necessarily give the same score to submissions with the same string**.

A submission is said to be an **original** submission if the string in the submission is never submitted in any earlier submission.
A submission is said to be the **best** submission if it is an original submission with the highest score. If there are multiple such submissions, only the earliest one is considered the best submission.

Find the index of the best submission.

Constraints

- $1 \leq N \leq 10^5$
- S_i is a string consisting of lowercase English characters.
- S_i has a length between 1 and 10, inclusive.
- $0 \leq T_i \leq 10^9$
- N and T_i are integers.

Input

Input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \\
S_1 \ T_1 \\
S_2 \ T_2 \\
\vdots \\
S_N \ T_N
\end{array}
$$

Output

Print the answer.

Sample 1

Input	Output
3 aaa 10 bbb 20 aaa 30	2

We will refer to the i-th earliest submission as Submission i.
Original submissions are Submissions 1 and 2. Submission 3 is not original because it has the same string as that in Submission 1.
Among the original submissions, Submission 2 has the highest score. Therefore, this is the best submission.

Sample 2

Input	Output
5 aaa 9 bbb 10 ccc 10 ddd 10 bbb 11	2

Original submissions are Submissions 1, 2, 3, and 4.
Among them, Submissions 2, 3, and 4 have the highest scores. In this case, the earliest submission among them, Submission 2, is the best.
As in this sample, beware that if multiple original submissions have the highest scores, only the one with the earliest among them is considered the best submission.

Sample 3

<table>
<tr><th>Input</th><th>Output</th></tr>
<tr><td><pre>10
bb 3
ba 1
aa 4
bb 1
ba 5
aa 9
aa 2
ab 6
bb 5
ab 3</pre></td><td><pre>8</pre></td></tr>
</table>

Problem H. Flip Edge

Time Limit 2000 ms

Problem Statement

You are given a directed graph with N vertices and M edges. The i-th edge $(1 \leq i \leq M)$ is a directed edge from vertex u_i to vertex v_i.

Initially, you are at vertex 1. You want to repeat the following operations until you reach vertex N:

- Perform one of the two operations below:
  - Move along a directed edge from your current vertex. This incurs a cost of 1. More precisely, if you are at vertex v, choose a vertex u such that there is a directed edge from v to u, and move to vertex u.
  - Reverse the direction of all edges. This incurs a cost of X. More precisely, if and only if there was a directed edge from v to u immediately before this operation, there is a directed edge from u to v immediately after this operation.

It is guaranteed that, for the given graph, you can reach vertex N from vertex 1 by repeating these operations.

Find the minimum total cost required to reach vertex N.

Constraints

- $2 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 2 \times 10^5$
- $1 \leq X \leq 10^9$
- $1 \leq u_i \leq N$ $(1 \leq i \leq M)$
- $1 \leq v_i \leq N$ $(1 \leq i \leq M)$
- For the given graph, it is guaranteed that you can reach vertex N from vertex 1 by the operations described.
- All input values are integers.

Input

The input is given from Standard Input in the following format:

$$
\begin{array}{l}
N \;\; M \;\; X \\
u_1 \;\; v_1 \\
u_2 \;\; v_2 \\
\vdots \\
u_M \;\; v_M
\end{array}
$$

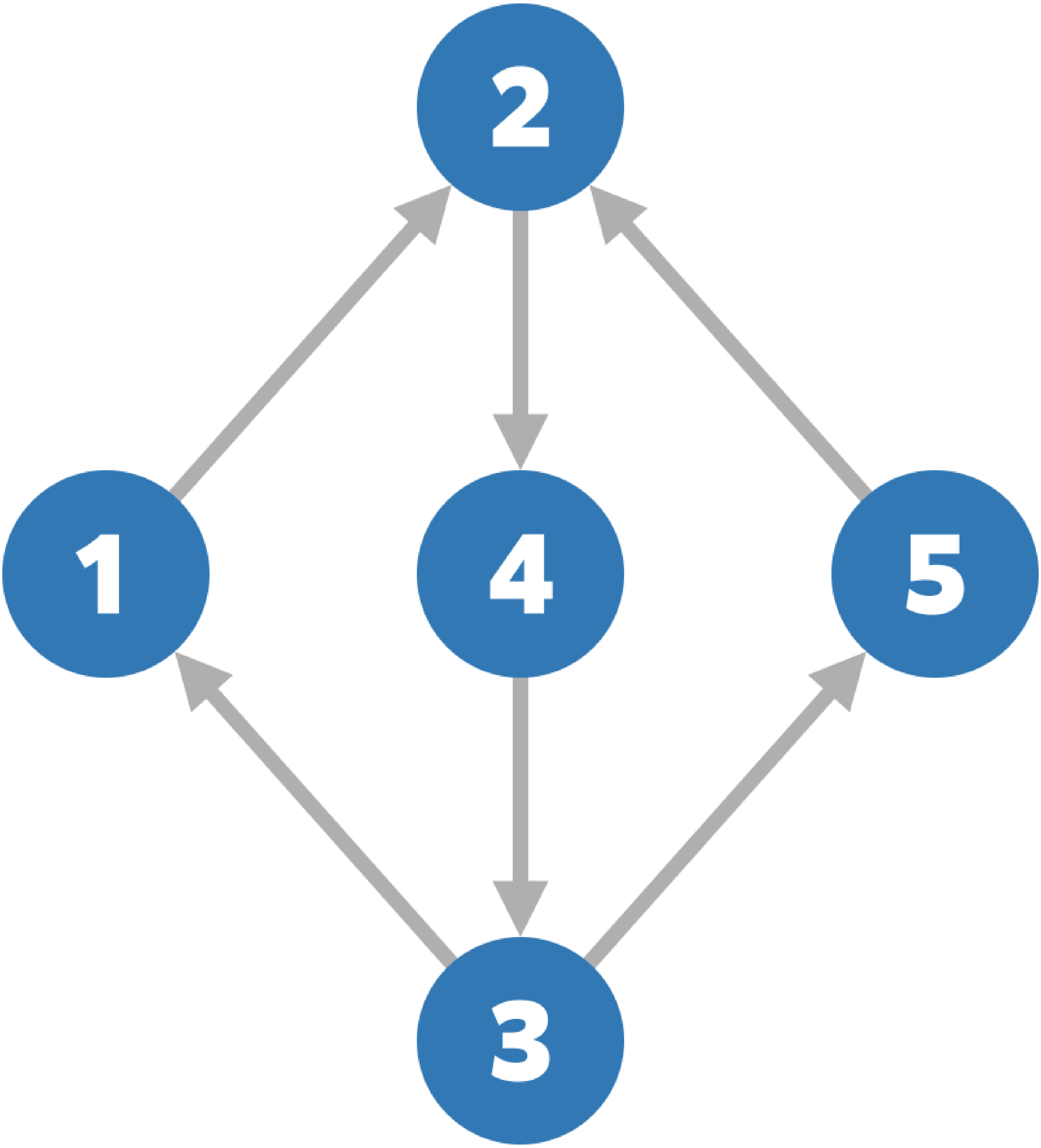
Output

Print the minimum total cost required to reach vertex N.

Sample 1

Input	Output
5 6 5 1 2 2 4 3 1 3 5 4 3 5 2	4

The given graph looks like this:



You can reach vertex 5 with a total cost of 4 by doing the following:

- Move to vertex 2 at a cost of 1.
- Move to vertex 4 at a cost of 1.
- Move to vertex 3 at a cost of 1.
- Move to vertex 5 at a cost of 1.

It is impossible to reach vertex 5 with a total cost of 3 or less, so print `4`.

Sample 2

Input	Output
5 6 1 1 2 2 4 3 1 3 5 4 3 5 2	3

The graph is the same as in Sample 1, but the cost to reverse edges is different.

You can reach vertex 5 with a total cost of 3 as follows:

- Move to vertex 2 at a cost of 1.
- Reverse all edges at a cost of 1.
- Move to vertex 5 at a cost of 1.

It is impossible to reach vertex 5 with a total cost of 2 or less, so print `3`.

Sample 3

Input	Output
8 7 613566756 2 1 2 3 4 3 4 5 6 5 6 7 8 7	4294967299

Note that the answer may exceed the 32-bit integer range.

Sample 4

Input	Output
20 13 5 1 3 14 18 18 17 12 19 3 5 4 6 13 9 8 5 14 2 20 18 8 14 4 9 14 8	21

Problem I. Shortest Routes I

Time Limit 1000 ms
Mem Limit 524288 kB

There are n cities and m flight connections between them. Your task is to determine the length of the shortest route from Syrjälä to every city.

Input

The first input line has two integers n and m: the number of cities and flight connections. The cities are numbered $1, 2, \dots, n$, and city 1 is Syrjälä.

After that, there are m lines describing the flight connections. Each line has three integers a, b and c: a flight begins at city a, ends at city b, and its length is c. Each flight is a one-way flight.

You can assume that it is possible to travel from Syrjälä to all other cities.

Output

Print n integers: the shortest route lengths from Syrjälä to cities $1, 2, \dots, n$.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input	Output
3 4 1 2 6 1 3 2 3 2 3 1 3 4	0 5 2

Problem J. Flight Discount

Time Limit 1000 ms
Mem Limit 524288 kB

Your task is to find a minimum-price flight route from Syrjälä to Metsälä. You have one discount coupon, using which you can halve the price of any single flight during the route. However, you can only use the coupon once.

When you use the discount coupon for a flight whose price is x, its price becomes $\lfloor x/2 \rfloor$ (it is rounded down to an integer).

Input

The first input line has two integers n and m: the number of cities and flight connections. The cities are numbered $1, 2, \ldots, n$. City 1 is Syrjälä, and city n is Metsälä.

After this there are m lines describing the flights. Each line has three integers a, b, and c: a flight begins at city a, ends at city b, and its price is c. Each flight is unidirectional.

You can assume that it is always possible to get from Syrjälä to Metsälä.

Output

Print one integer: the price of the cheapest route from Syrjälä to Metsälä.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$
- $1 \leq c \leq 10^9$

Example

Input	Output
3 4 1 2 3 2 3 1 1 3 7 2 1 5	2